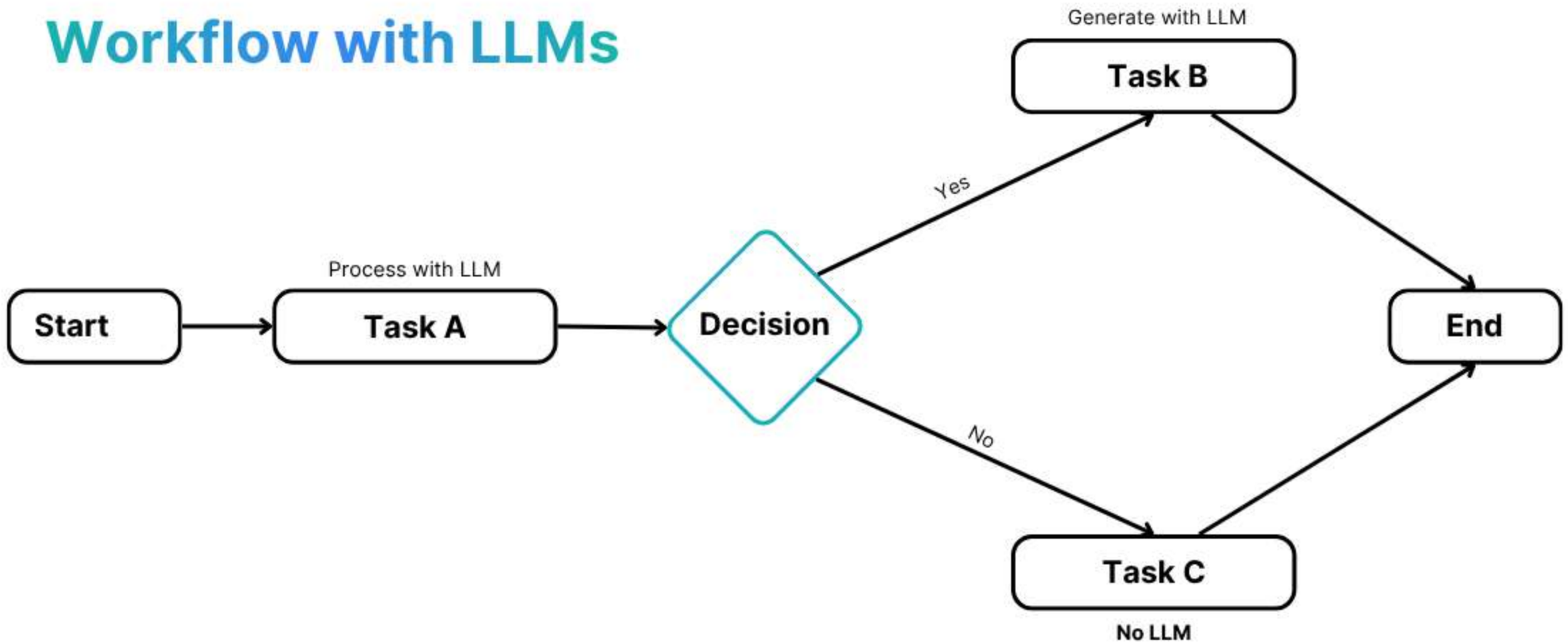


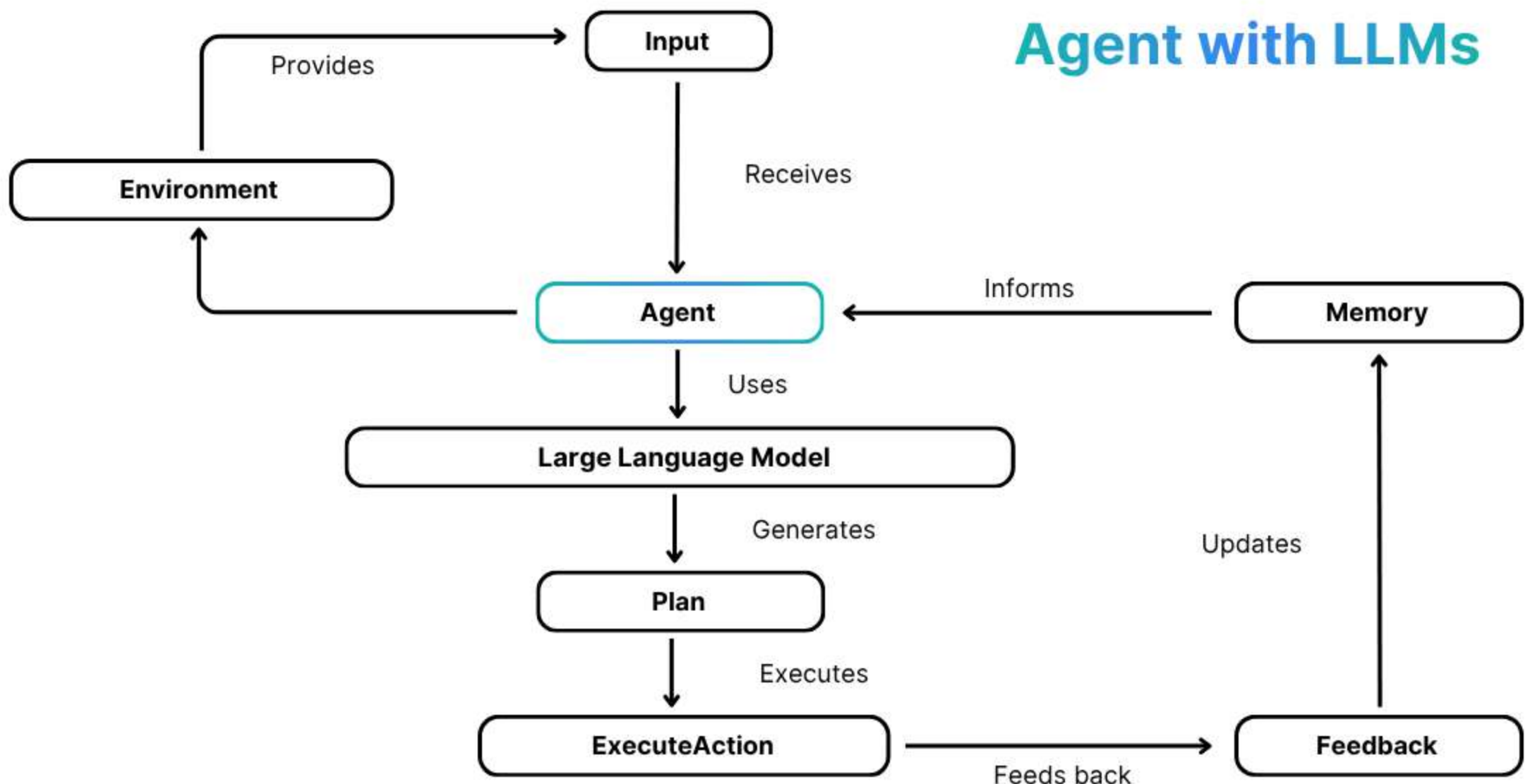
Workflows vs Agents

When to use each and everything in between

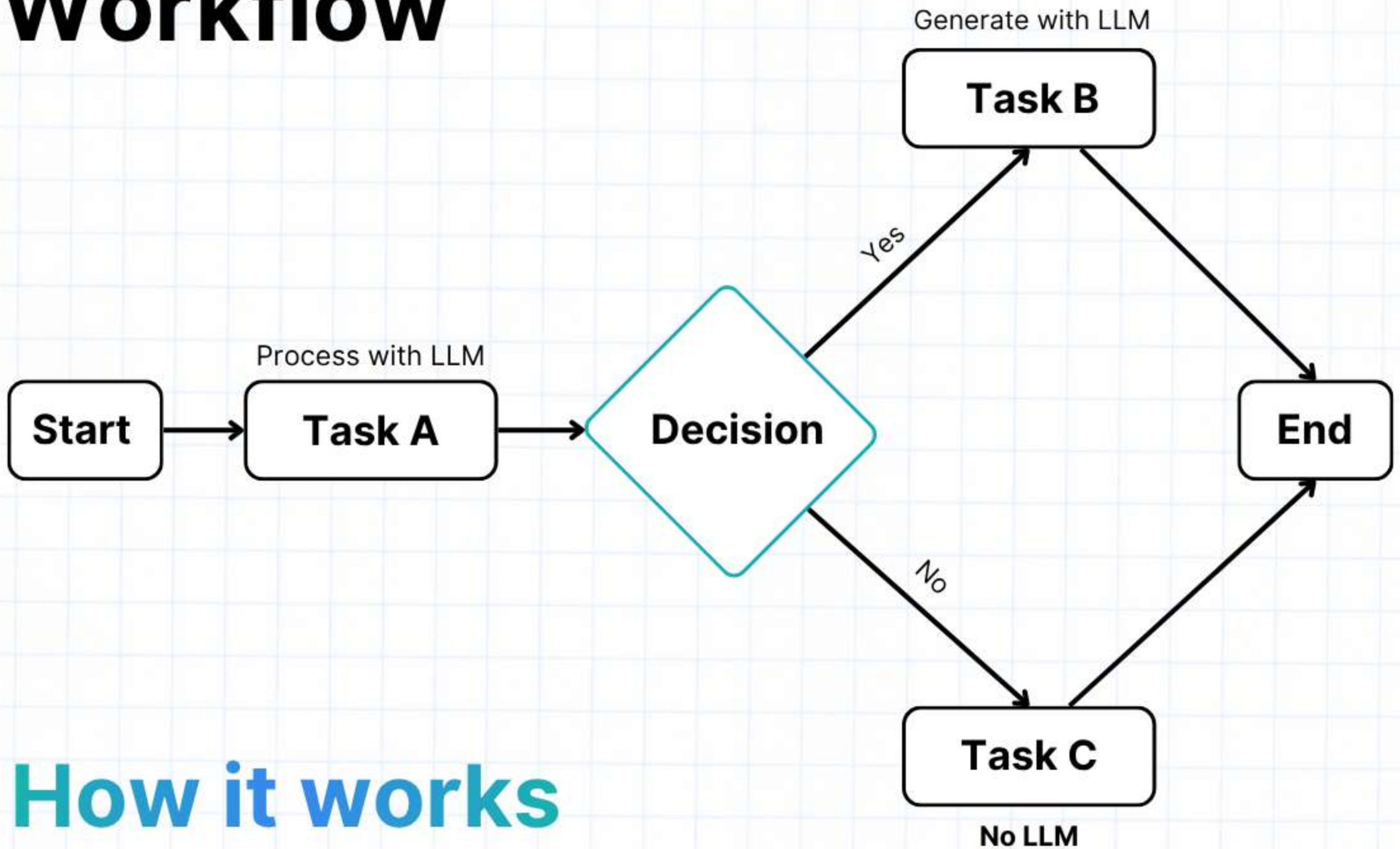
Workflow with LLMs



Agent with LLMs



Workflow



How it works

Deterministic

Same input always gives the same output type. No surprises.

Developer-controlled

Every step, branch, and LLM call is designed upfront by you.

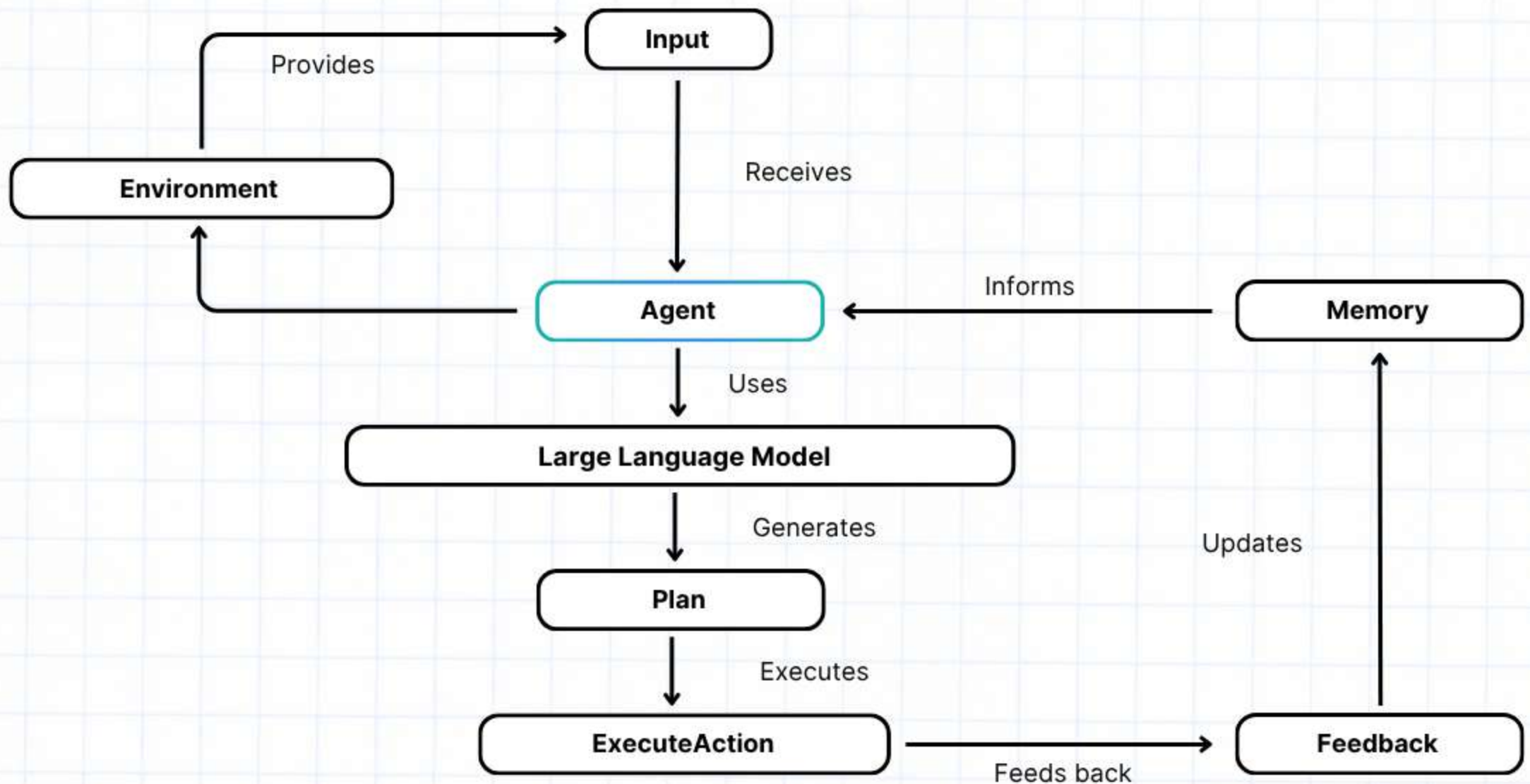
Easy to debug

When something breaks, you know exactly where to look.

Rigid

Cannot handle situations the designer didn't anticipate.

Agent



How it works

Autonomous

The LLM decides which tool to call and when to stop

Adaptive

Handles new situations it was never explicitly programmed for.

Harder to control

Outputs vary. Errors can compound across steps.

Powerful

Can solve complex tasks through dynamic reasoning and loops.

Key Differences

Workflow		Agent	
Flexibility			
Rigid: follows the same path every time		Flexible: adapts to new and unexpected situations	
Control			
Full: developer defines every step		Partial: LLM decides the path at runtime	
Debuggability			
Easy: every step is logged and traceable		Hard: decisions are dynamic and can compound	
Cost			
Lower: fewer LLM calls, predictable spend		Higher: multi-step loops, tool calls add up	
Risk			
Low: failures are isolated and predictable		Higher: errors can snowball across steps	

When to use which?

Many 'Agents' are actually Workflows in disguise.

Start simple. Add autonomy only when the task genuinely needs it.

Use a Workflow when...

- Steps are known and fixed upfront
- Compliance and auditability matter
- Task runs repeatedly with same logic
- Speed and low cost are priorities
- Errors must be caught and isolated

e.g. Invoice processing, scheduled reports e.g. Invoice processing, email campaigns

Use an Agent when...

- Task structure is unknown upfront
- Novel or unpredictable situations arise
- Multiple tools need to be chained dynamically
- Goal matters more than the exact path
- Real-time adaptation is essential

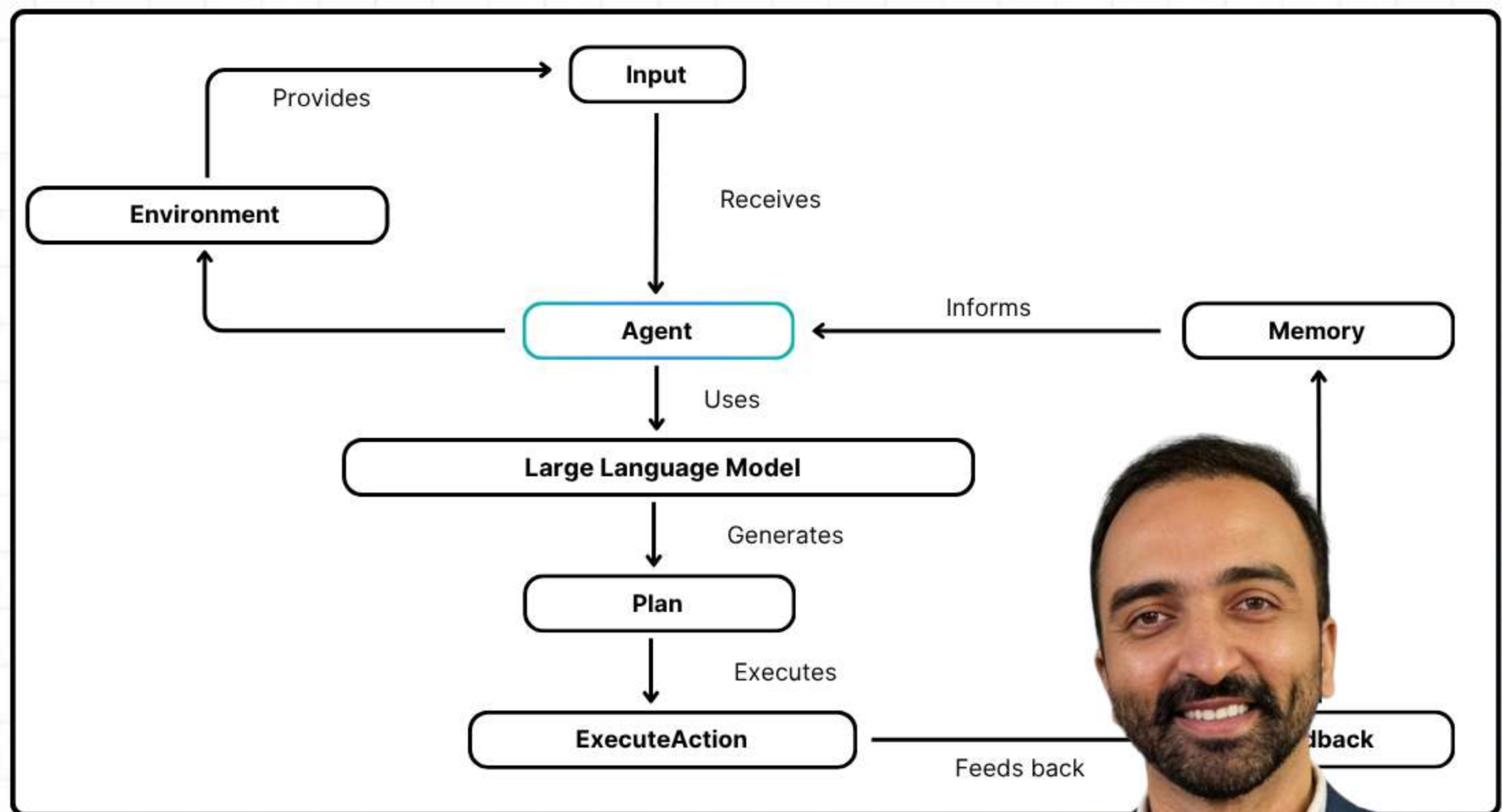
e.g. Customer support, research, market analysis

The Golden Rule

Start with the simplest architecture that could work. Reach for agents only when workflows have a clear ceiling.

Go Deeper on Hack Session at DataHack Summit 2026

Workflows, Agents, and Everything In Between



Manu Joseph

Principal Data Scientist

Walmart  Global Tech India

